

Jet engine cycle analysis

Group 10 - integrated model and validation

The NASA-property model predicts an exhaust speed of **778.83 m/s**, a combustor outlet temperature of **1082.46 K**, and matched compressor/turbine power of **33.242 MW**. All 21 independent numerical residual checks pass.

Inputs and provisional settings

Group input	Value	Model setting	Value
Fuel	H2	eta_c / eta_t / eta_n	1 / 1 / 1
Ambient T / P	300 K / 100 kPa	Fuel temperature	298.15 K
Flight speed	200 m/s	P4 / P3	1
Compressor ratio	9	Combustor heat loss	0 W
Fuel flow / AF	0.58 kg/s / 204.42	Shaft	Lossless

Settings reminder: the efficiencies, fuel inlet temperature and loss assumptions are provisional model inputs, not confirmed against the current course settings. They are centralized in the code and exported in `provisional_assumptions.csv`.

The model is steady and adiabatic by default, with negligible potential-energy changes and negligible internal kinetic energy at states 2-5. Air is 21% O₂ and 79% N₂ by mole. Hydrogen burns completely; downstream composition is frozen. The nozzle exit pressure is prescribed as ambient.

Complete state results

State	T [K]	P [kPa]	v [m/s]	h [kJ/kg]	s_mix [kJ/kg K]
1	300.00	100.00	200.00	1.91	6.89530
2	319.78	125.11	0.00	21.91	6.89530
3	591.58	1125.98	0.00	302.28	6.89530
4	1082.46	1125.98	0.00	300.81	7.83300
5	848.87	423.74	0.00	21.80	7.83300
6	580.54	100.00	778.83	-281.49	7.83300

States: 1 inlet; 2 diffuser outlet; 3 compressor outlet; 4 combustor outlet; 5 turbine outlet; 6 nozzle exit. States 1-3 use air properties; states 4-6 use product properties. Enthalpy and mixture entropy above are evaluated directly from NASA polynomials at the solved temperatures.

Air flow = **118.5636 kg/s**; fuel flow = **0.5800 kg/s**; product flow = **119.1436 kg/s**. Internal zero velocities represent neglected kinetic-energy terms, not zero mass flow.

Thermodynamic method

NASA species enthalpies include formation enthalpies. Mixture properties are mass-weighted: $h(T) = \sum(Y_i h_i(T))$; $s_T(T) = \sum(Y_i S_{\text{Nasa}_i}(T))$; $R_{\text{mix}} = R_u/M_{\text{mix}}$. The main script inverts property curves on a 200-3000 K grid with 1 K spacing. No constant-cp Poisson relation is used. [1-3]

Diffuser and compressor - Part 1

```
h2 = h1 + v1^2/2                      (v2 = 0)
P2 = P1 exp[(s_T,2 - s_T,1)/R_air]
s_T,3s = s_T,2 + R_air ln(P3/P2)
h3 = h2 + (h3s - h2)/eta_c
Wcomp = m_air (h3 - h2)
```

The diffuser converts flight kinetic energy into enthalpy and satisfies $s_2=s_1$. Invert $h_{\text{air}}(T_2)=h_2$ to find T_2 . For the compressor, entropy gives T_{3s} at the prescribed pressure ratio; efficiency gives actual h_3 , then inversion gives T_3 . Here $T_3=591.58$ K and $T_{3s}=591.58$ K.

Combustion and combustor - Part 2

```
H2 + 0.5 O2 -> H2O
n_products = n_in + [-1 -0.5 0 1 0] n_H2,in
h4 = (m_air h3 + m_fuel h_fuel - Qloss)/m_products
h_products(T4) = h4;                P4 = (P4/P3) P3
```

Species order is [H2, O2, CO2, H2O, N2]. Mass flows are converted to molar flows with database molecular masses; product mass and mole fractions follow from the reacted flows. The equivalence ratio is 0.1667, leaving excess oxygen. Adding an LHV term would double-count chemical energy. This is an open-flow enthalpy balance, unlike the closed-vessel internal-energy exercise in Lecture 1. [2,4]

Turbine and nozzle - Part 3

```
h5 = h4 - Wcomp/m_products
h5s = h4 - (h4 - h5)/eta_t
P5 = P4 exp[(s_T,5s - s_T,4)/R_products]
s_T,6s = s_T,5 + R_products ln(Pamb/P5)
h6 = h5 - eta_n (h5 - h6s)
v6 = sqrt[2(h5 - h6)]                (v5 = 0)
```

The turbine supplies exactly the compressor power through a lossless shaft. Its mass flow includes added fuel. Inversions yield T_5 , T_{5s} and T_{6s} ; turbine pressure follows from $s_{5s}=s_4$. The nozzle converts the remaining enthalpy drop into kinetic energy while expanding to ambient. Enthalpies inside the velocity formula are in J/kg. [3,5]

Independent validation - Part 4

Each solved state is re-evaluated directly with HNasa and SNasa. This checks the temperature inversions rather than only substituting an enthalpy target back into its defining equation. The table reports signed residuals; PASS requires absolute residual no greater than the tolerance.

Check	Residual	Tolerance	Unit
Diffuser energy	0.009484	2	J/kg
Compressor power	-1.936	237.1	W
Combustor energy	-0.3066	237.7	W
Turbine power	-0.9184	238.3	W
Shaft power	1.018	475.4	W
Nozzle energy	0.0122	2	J/kg
Whole engine energy	3.289	237.7	W
Combustion mass	-1.588e-14	1.191e-07	kg/s
Element H	0	5.754e-07	mol atoms/s
Element O	-2.274e-13	1.726e-06	mol atoms/s
Element C	0	1e-09	mol atoms/s
Element N	0	6.493e-06	mol atoms/s
Air X sum	0	1e-10	-
Air Y sum	0	1e-10	-
Product X sum	0	1e-10	-
Product Y sum	0	1e-10	-
Max h inversion	0.0276	1	J/kg
Diffuser isentropy	0.0008214	0.01	J/(kg K)
Compressor reference isentropy	-0.0004973	0.01	J/(kg K)
Turbine reference isentropy	-3.352e-05	0.01	J/(kg K)
Nozzle reference isentropy	0.0002819	0.01	J/(kg K)

A 1 J/kg single-state enthalpy allowance and 0.01 J/(kg K) entropy allowance account for interpolation. Flow-energy tolerances sum allowed state errors multiplied by mass flow. The largest enthalpy inversion error is only **0.02760 J/kg**. The largest tolerance fraction is 0.08214 (limit 1). Reconstructed component efficiencies also pass. These checks assess numerical consistency, not physical model accuracy.

The whole-engine balance is $m_{\text{air}}(h_1 + v_1^2/2) + m_{\text{fuel}} h_{\text{fuel}} - m_{\text{products}}(h_6 + v_6^2/2) - Q_{\text{loss}} = 0$. Its **3.289 W** residual is tiny compared with the energy flows. Internal compressor/turbine work cancels for the lossless shaft.

Separate solver: verify_group10.m solves the same run using direct NASA functions and fzero. Maximum temperature disagreement is **0.000445 K**; exhaust-speed disagreement is **0.000662 m/s**. The check uses different numerical inversion; it shares the same physical assumptions and database.

Results interpretation

Group 10 - provisional settings

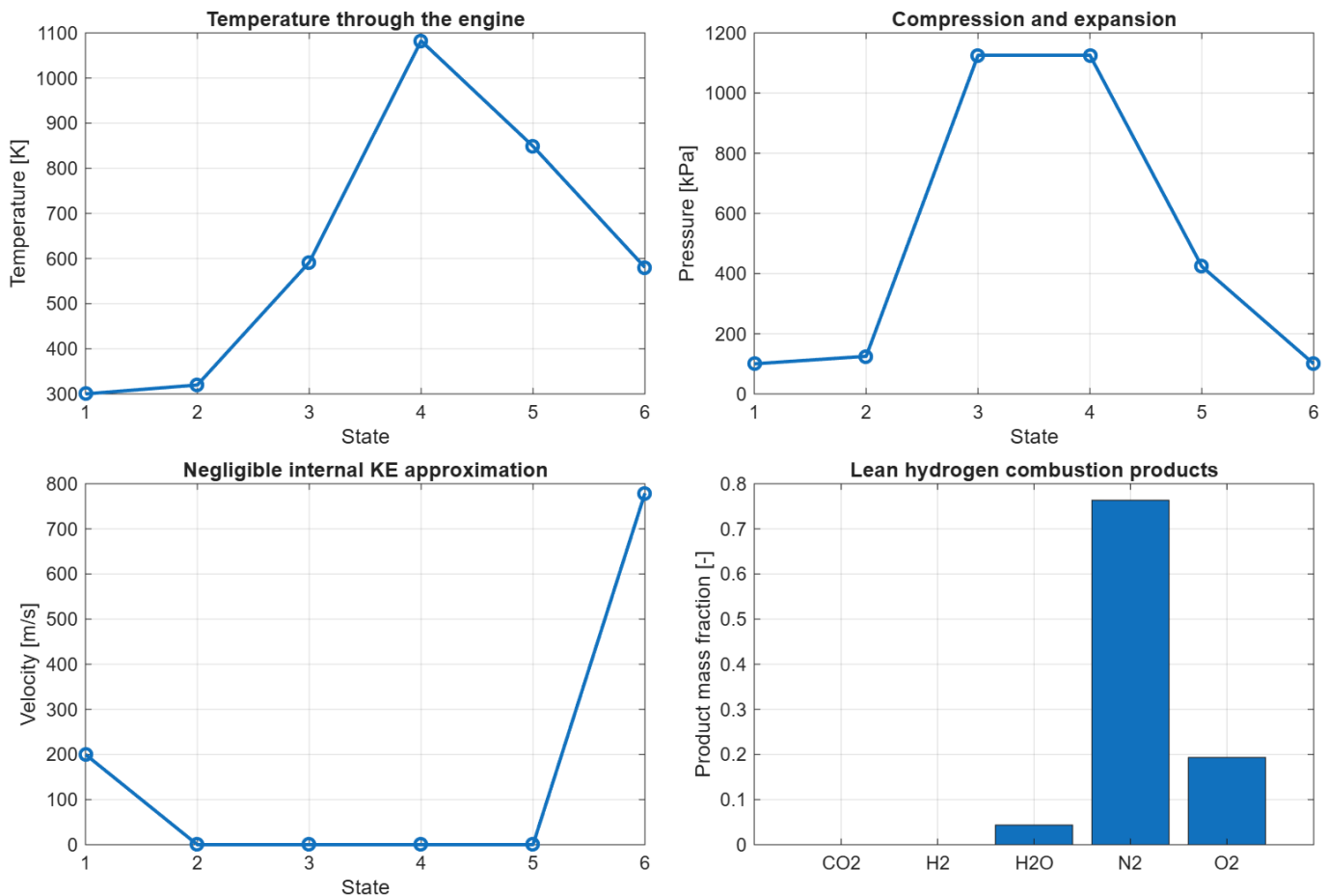


Figure 1. Computed state trends and product composition. Connecting lines guide the eye; they are not spatially resolved profiles or equilibrium paths.

What the trends mean

The diffuser and compressor heat the incoming air. Combustion produces the highest temperature. The turbine then extracts compressor power, and the nozzle accelerates the products. A local frozen-composition sound-speed calculation gives an exit Mach number of **1.6023**.

The prescribed fully expanded solution is supersonic and assumes suitable nozzle geometry; this is not a prediction for an arbitrary converging-only nozzle.

The combustor can have h_4 slightly below h_3 even though T_4 is much higher. Enthalpy flow is conserved while composition changes the enthalpy-temperature curve. Likewise, $h_6 = -281.49$ kJ/kg is valid: NASA formation-enthalpy reference values can be negative. The positive $h_5 - h_6$ drives nozzle acceleration.

Entropy and composition

The original model omits the constant mixing contribution for each fixed mixture. This leaves its isentropic calculations valid. Part 4 also reports full mixture entropy using species partial pressures:
 $s_{\text{mix}} = s_{\text{model}} - \sum [Y_i (R_u / M_i) \ln(X_i)]$, excluding zero-fraction species. Do not interpret $s_4 - s_3$ alone as combustor entropy production; fuel entropy and the unequal stream mass flows must also be included.

Product mass fractions [H₂, O₂, CO₂, H₂O, N₂]: [0.000000, 0.193148, 0.000000, 0.043503, 0.763348]. Mole fractions: [0.000000, 0.169077, 0.000000, 0.067641, 0.763282]. Complete species flows are exported to composition.csv.

Integration, review and source record

Assessment of the existing work

Part 1: correct NASA-based diffuser/compressor method for the chosen baseline. The original current-folder dependency is fixed; direct property checks strengthen its validation. **Part 2:** correct hydrogen stoichiometry, mass/element accounting and formation-enthalpy energy balance. **Part 3:** correct turbine mass-flow factor, efficiency equations and nozzle energy conversion. Default efficiencies and losses remain provisional. No grade is inferred from the rubric.

Part 4 implementation and reproducibility

The model remains a single sequential script, with one input/settings block and script-relative database access. It collects all six stations, evaluates 21 residual checks, enforces physical trends and valid efficiencies, and exports the state, composition, validation and assumptions tables, a MAT result file and the figure. The component comparison and a plain-text results summary are also exported. Failed residual checks are saved with their names for diagnosis. The supplied NASA base functions and database remain unchanged.

```
JetEngine_Group10      % solve, validate and export results
verify_group10         % compare against direct NASA roots
```

Extract the scripts package and use that folder as MATLAB Current Folder, or add it to the MATLAB path. Outputs are regenerated under results/. MATLAB R2026a was used for validation. PART4_EXPLANATION.md gives the detailed integration and checking rationale.

Items to confirm before submission

Confirm the default efficiencies, fuel inlet state, combustor losses, shaft model and any turbine-temperature limit against the current course files. Use the actual Canvas Word template, fill in student details and verify packaging/team requirements there. This report supplies technical content; its structure has not been checked against the unavailable template. The 2026 Lecture 2, p. 15, gives 9 October regular and 16 October late submission, with the late grade capped at 8. The 2025 handout dates are superseded by that supplied lecture.

Sources inspected

[1] 4EB00 Special Topic Jet Engine Info 2025.pdf, pp. 1-2: NASA/MATLAB method and deliverables; 4EB00 Special Topic Jet Engine Rubric.pdf, pp. 1-2: methodology, composition and result criteria.

[2] Lecture 1 Ideal Gas Mixtures 2026.pdf, pp. 9-10, 19-27, 29-34: mixture definitions, formation enthalpy, NASA functions and reacting systems.

[3] Lecture 2 Cycle analysis 2026.pdf, pp. 7-15: component models, entropy method, group conditions and 2026 dates.

[4] CollegeThermo1-2-eng-1.pdf: systems, properties, first law and enthalpy. [5] CollegeThermo3-4-eng.pdf, pp. 9-16: steady-flow energy balances and turbine/nozzle models.

[6] CollegeThermo5-6-eng.pdf: heat-engine cycles and efficiency context. Its constant-cp closed Brayton-cycle formula is not used for this reacting turbojet.

[7] Jet_Engine_Project_Working_Plan_Group10.docx, sections 3-6: four-part scope, integration and validation. The working plan defines the four-part division.

[8] github.com/kapnistos/Thermodynamics, commit 8bfbf18a1d044f59f87f9d72e3682f26f9f8e72a, downloaded 26 September 2026: main script, README, starter, scratch file, all General functions/database, older lectures and working plan.